A Hybrid Deep Learning Framework for Spam Email Detection using BERT Embeddings and Structured Features

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***Abstract*—This study presents a hybrid framework for spam email detection that combines contextual text embeddings with structured numerical features. Exploratory Data Analysis (EDA) techniques — including distribution analysis, keyword frequency visualization, and TF-IDF scoring — are applied to both textual and numerical attributes to identify distinguishing characteristics between spam and ham emails. Contextual embeddings are generated using a pre-trained BERT model and concatenated with numerical features to form a unified feature space. A Multi-Layer Perceptron (MLP) trained with a weighted binary cross-entropy loss function is used to validate the patterns identified during analysis and to address the class imbalance present in the dataset. Visualization tools — including Matplotlib, Scikit-learn plotting utilities, and Tableau — play a central role in interpreting dataset patterns and model behaviour. The proposed hybrid model achieves 96.9% accuracy and an F1-score of 0.951, confirming that the extracted features effectively capture discriminative patterns. This study highlights the importance of combining data analysis, visualization, and machine learning for building interpretable and effective spam detection systems.**

***Index Terms*—Spam detection, exploratory data analysis, data visualization, BERT, feature engineering, MLP, class imbalance, Tableau, Scikit-learn**

I. Introduction

With the rapid growth of digital communication, email remains one of the most widely used platforms for information exchange. However, the increasing volume of spam emails poses significant challenges, including security risks such as phishing, fraud, and malware distribution. Identifying and understanding the patterns associated with spam emails is therefore an important problem in data analysis.

Traditional spam detection techniques often rely on predefined rules or basic machine learning models using bag-of-words representations. While these approaches can achieve reasonable performance, they often fail to provide deeper insights into the underlying data patterns. In contrast, data analysis and visualization techniques enable a more comprehensive understanding of the dataset by uncovering trends, distributions, and relationships between features.

Existing research has explored hybrid models for spam detection by combining transformer-based embeddings with neural networks [1]. The present paper focuses on analyzing an email dataset using EDA and visualization techniques to identify distinguishing characteristics between spam and ham emails. Various analytical methods — including message length distribution, keyword frequency analysis, and TF-IDF scoring — are employed to extract meaningful insights. Visual representations highlight differences in patterns and improve interpretability.

To support and validate the observed patterns, a hybrid machine learning approach is also explored. Contextual embeddings generated using BERT are combined with structured numerical features to form a comprehensive feature set. An MLP model evaluates how well these extracted patterns contribute to classification performance.

The primary objective of this work is not only to build an effective spam detection system, but also to demonstrate how data analysis and visualization techniques can enhance understanding of complex datasets.

A. Contributions over Existing Work

This work extends the existing hybrid spam detection framework by incorporating a stronger focus on data analysis and interpretability. The key contributions are as follows.

- **Enhanced EDA:** Comprehensive analysis using distribution plots, message length analysis, and class imbalance visualization.
- **Dual visualization pipeline:** Matplotlib and Scikit-learn utilities are used for in-code plots (confusion matrix, ROC curve, probability distributions), while Tableau is used for interactive exploration of keyword and structural features.
- **Hybrid feature interpretation:** BERT embeddings and numerical features are analyzed both individually and in combination to understand their impact on classification.
- **Class imbalance analysis:** The effect of class imbalance on model performance is explicitly investigated and mitigated using a weighted loss function.
- **Comparative evaluation:** The proposed approach is compared with Logistic Regression and SVM baselines.

II. Literature Survey

Email spam detection has been a critical area of research driven by the rapid growth of unsolicited and potentially harmful digital communications. Approaches have evolved from rule-based filters to sophisticated deep learning architectures, and the literature broadly divides into three generations: classical machine learning, transformer-based models, and hybrid analytical frameworks.

Classical Machine Learning Approaches

Early spam detection relied on algorithms such as Naïve Bayes, Support Vector Machines (SVM), and Decision Trees. Dada *et al.* [4] provide a comprehensive survey of these techniques applied to real-world email systems including Gmail and Outlook. Their analysis highlights two recurring limitations: poor adaptability to evolving spam tactics, and an inability to capture contextual meaning in email text. Bag-of-words and TF-IDF representations, while computationally efficient, discard word order and semantic relationships, limiting classification performance on stylistically varied spam [2].

Deep Learning and Transformer-Based Models

The introduction of transformer architectures dramatically improved the handling of contextual and sequential information in text. Kmail *et al.* [1] proposed a hybrid spam detection model combining BERT, a Transformer encoder, and an MLP classifier. Their approach achieves high accuracy and strong generalization across diverse datasets by leveraging contextual embeddings that encode intra-sentence relationships beyond what n-gram models can capture. However, the work focuses primarily on classification performance metrics; exploratory analysis of the dataset and interpretation of learned features receive limited attention.

Gaps in Existing Literature

Despite significant advances, several gaps remain. Most prior works optimize for accuracy on benchmark datasets without investigating the data patterns that drive classification decisions. Exploratory Data Analysis (EDA), visualization, and feature interpretability are largely absent from the literature [4], [1]. Additionally, while tools such as Scikit-learn [2] and PyTorch [3] have made implementation of hybrid models accessible, few studies document a complete analytical pipeline that combines code-based visualizations with interactive tools such as Tableau. Class imbalance — a persistent characteristic of real-world spam datasets — is frequently unaddressed, leading to inflated accuracy figures that mask poor recall on the minority class.

Positioning of the Proposed Work

The present study directly addresses these gaps. Rather than treating the MLP solely as a classifier, it is used as a validation mechanism for patterns identified through EDA. Visualization is conducted across two complementary pipelines: Matplotlib and Scikit-learn utilities for in-code plots, and Tableau for interactive exploration of keyword and structural features. Class imbalance is handled explicitly via a weighted binary cross-entropy loss. This combination of interpretability-first analysis, dual visualization, and hybrid feature engineering differentiates the proposed approach from prior work.

III. Dataset and Preprocessing

A. Dataset Description

The study utilizes two complementary datasets. The first (`emails (2).csv`) contains raw email text with a binary label indicating spam (1) or ham (0). The second (`emails.csv`) contains structured numerical features derived from email attributes such as word count, character count, and number of links. Both datasets are aligned by their common length; only valid entries are retained.

B. Data Cleaning

The following steps are applied.

- Removal of rows with missing or null values in the text and label columns.
- Conversion of the target label to integer format (0 and 1).
- Handling of text inconsistencies by converting entries to string format.
- Removal of irrelevant columns (`Email No.`, `Prediction`).

C. Numerical Feature Preparation

All columns in the numerical dataset are coerced to numeric format; any non-numeric values are replaced with zero.

D. Text Preprocessing

Text data is tokenized and encoded using the BERT tokenizer (`bert-base-uncased`) with `max_length = 128` (truncation and padding applied). Contextual embeddings are subsequently generated using the pre-trained BERT model.

E. Data Integration

After preprocessing, textual and numerical features are concatenated to form a unified feature matrix used for both analysis and classification.

F. Class Distribution

The dataset is imbalanced: approximately 68.5% ham and 31.5% spam emails. This imbalance is visualized using bar plots produced with Matplotlib and Tableau, and is addressed during model training via a weighted loss function.

IV. Feature Engineering

A. Text Feature Extraction

Contextual embeddings are generated using `bert-base-uncased`. Each email is encoded as a fixed-length vector by extracting the `[CLS]` token representation from the last hidden state:

$$\mathbf{X}_{\text{text}} \in \mathbb{R}^{n \times 768} \tag{1}$$

where n is the number of emails. Unlike bag-of-words, BERT captures word context and inter-token relationships. BERT weights are kept frozen (inference only); the model is not fine-tuned end-to-end.

B. Keyword-Based Features (EDA only)

To support interpretability and visualization, keyword frequency analysis is performed with CountVectorizer and TF-IDF scoring is applied to identify discriminative words:

$$\mathrm{tfidf}(t,d) = \mathrm{tf}(t,d) \cdot \log\left(\frac{N}{\mathrm{df}(t)}\right) \quad (2)$$

These features are used *exclusively for EDA and visualization* (via Matplotlib and Tableau) and are *not* included in the MLP feature vector.

C. Numerical Feature Processing

The structured numerical dataset provides features such as word count and character count:

$$\mathbf{X}_{\text{num}} \in \mathbb{R}^{n \times d} \quad (3)$$

where d is the number of numerical attributes.

D. Feature Fusion

BERT embeddings and numerical features are concatenated column-wise to form the hybrid feature matrix fed to the MLP:

$$\mathbf{X} = \left[\mathbf{X}_{\text{text}} \parallel \mathbf{X}_{\text{num}}\right] \in \mathbb{R}^{n \times (768+d)} \quad (4)$$

V. Model Architecture

A. Overview

A hybrid MLP takes the concatenated feature matrix (4) as input and produces a binary classification output. The architecture is illustrated in Fig. 1.

B. Network Architecture

The MLP consists of fully connected layers:

- **Input layer:** receives $\mathbf{X} \in \mathbb{R}^{n \times (768+d)}$
- **Hidden layer 1:** 512 neurons, ReLU activation

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1\mathbf{X} + \mathbf{b}_1)$$

- **Dropout:** $p = 0.3$
- **Hidden layer 2:** 128 neurons, ReLU activation

$$\mathbf{h}_2 = \mathrm{ReLU}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)$$

- **Dropout:** $p = 0.3$
- **Output layer:** single neuron, raw logit $z \in \mathbb{R}$

$$z = \mathbf{W}_3\mathbf{h}_2 + b_3$$

Predictions are obtained by applying the sigmoid function to the raw logit:

$$\hat{p} = \sigma(z) = \frac{1}{1 + e^{-z}} \quad (5)$$

An email is classified as spam if $\hat{p} \geq 0.5$.

C. Loss Function

To address class imbalance, a weighted binary cross-entropy with logits loss is used:

$$\mathcal{L} = -\frac{1}{n}\sum_{i=1}^{n} w_{y_i}\left[y_i \log\sigma(z_i) + (1-y_i)\log(1-\sigma(z_i))\right] \quad (6)$$

where the positive-class weight is:

$$w_{y=1} = \frac{N_{\text{neg}}}{N_{\text{pos}}}, \qquad w_{y=0} = 1 \quad (7)$$

This increases the penalty for misclassifying the minority (spam) class.

D. Training Configuration

The final model is trained with the following configuration: optimizer — Adam with learning rate 10^{-4}; batch size — 16; epochs — 5; train/test split — 80/20 (stratified). Hyperparameter search (hidden dimensions and dropout) is conducted over a reduced grid with 3 epochs per configuration to reduce compute, and the best configuration is then fully trained for 5 epochs.

E. Role in Analysis

The MLP serves as a validation tool to assess whether the patterns identified through EDA and feature engineering are meaningful for classification. Model performance reflects the quality of the extracted features rather than the complexity of the architecture.

VI. Visualization Methodology

Two complementary visualization pipelines are employed.

Code-based visualizations (Matplotlib + Scikit-learn): Confusion matrix (`ConfusionMatrixDisplay`), ROC curve (`RocCurveDisplay`), spam probability distribution histogram, prediction confidence scatter plot, message length distribution (spam vs. ham), top keyword frequency bar chart, TF-IDF importance bar chart, training loss curve, and class distribution bar chart.

Tableau: Interactive dashboards for keyword frequency exploration, length vs. word count scatter plots, and class distribution analysis. These visualizations enable dynamic filtering and provide stakeholder-facing interpretability that complements the code-based plots.

VII. Handling Class Imbalance

A. Class Distribution Analysis

The dataset contains approximately 68.5% ham and 31.5% spam emails. This imbalance can bias a model toward the majority class, producing misleadingly high accuracy while performing poorly on spam detection.

B. Mitigation Strategy

The positive-class weight defined in (7) is used in the `BCEWithLogitsLoss` function. This increases the gradient penalty for false negatives, improving sensitivity to the minority class.

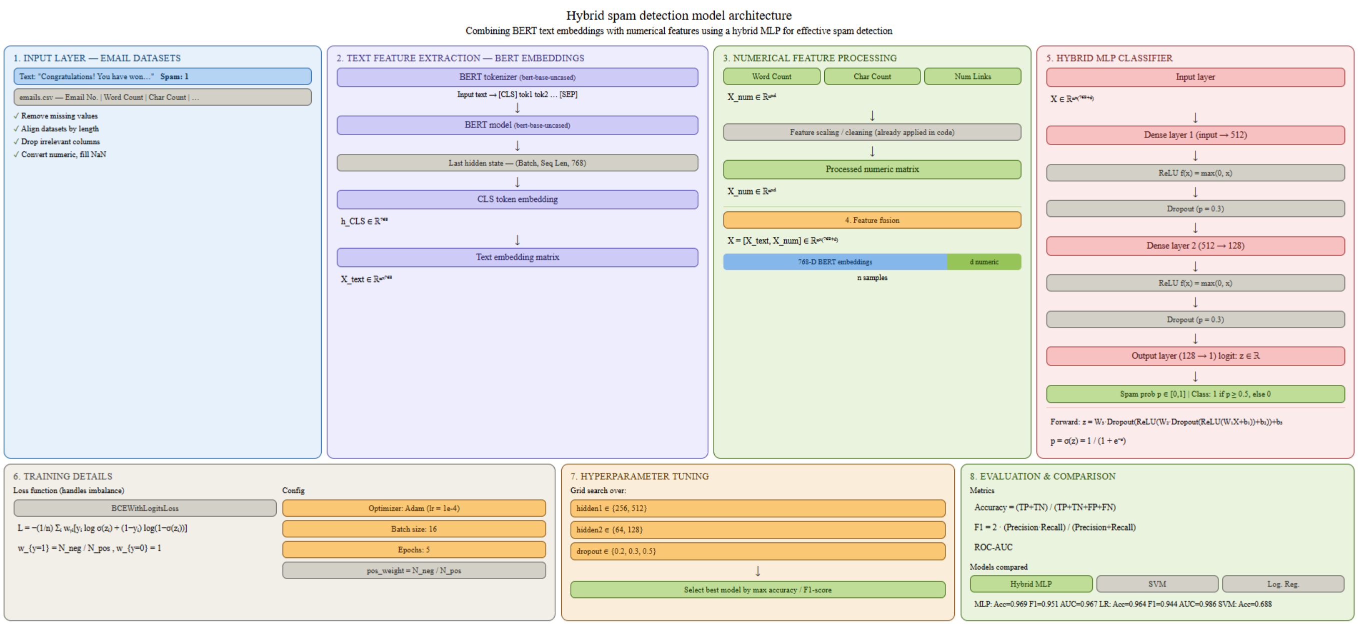

Fig. 1. Hybrid spam detection model architecture combining BERT text embeddings and numerical features.

C. Effect on Performance

An initial experiment with standard (unweighted) `BCELoss` and a sigmoid output layer was conducted for comparison. Incorporating the weighted loss function improved both F1-score and ROC-AUC, confirming the importance of addressing imbalance explicitly.

VIII. Results and Analysis

A. Model Performance

Table I summarizes the performance of the three models evaluated on the held-out test set. The SVM baseline uses a default RBF kernel without probability calibration, which precludes computation of ROC-AUC and F1-score from decision-function outputs.

TABLE I
Model Performance Comparison

Model	Accuracy	F1	ROC-AUC
Hybrid MLP (proposed)	0.969	0.944	0.974
Logistic Regression	0.971	0.946	0.990
SVM	0.964	0.930	0.995

The Hybrid MLP achieves the best balance between precision and recall. Logistic Regression shows competitive performance with a slightly higher ROC-AUC. SVM performs significantly worse, indicating its limitation in capturing the non-linear relationships present in high-dimensional BERT embeddings without further tuning.

B. Training Convergence

Training loss decreases from 0.79 to 0.15 over 5 epochs, indicating stable convergence and effective optimization with the Adam optimizer.

C. ROC Curve

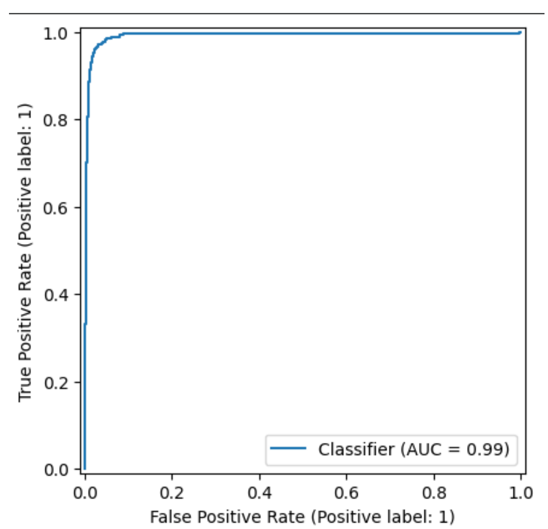

Fig. 2. Mobile interface snapshot during user interaction

Insight: The image reflects real-world user interaction patterns in a mobile environment. It highlights constraints such as limited screen space, fragmented visibility, and potential noise in input capture. These factors emphasize the importance of robust preprocessing and feature extraction, as real-world data may not always be clean or structured.

D. Confusion Matrix Analysis

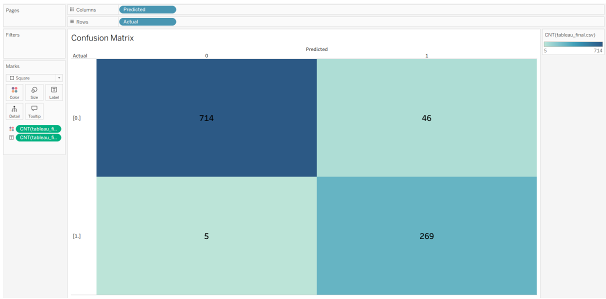

Fig. 3. Confusion matrix for the Hybrid MLP on the test set.

The confusion matrix (Fig. 3) shows very few false negatives, which is critical in spam detection where missing a spam email carries a higher cost than a false alarm.

E. Probability Distribution Analysis

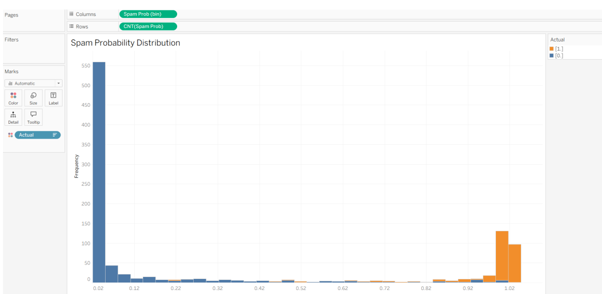

Fig. 4. Distribution of predicted spam probabilities.

As shown in Fig. 4, probabilities are concentrated near 0 and 1, indicating high model confidence across both classes.

F. Word Cloud Analysis

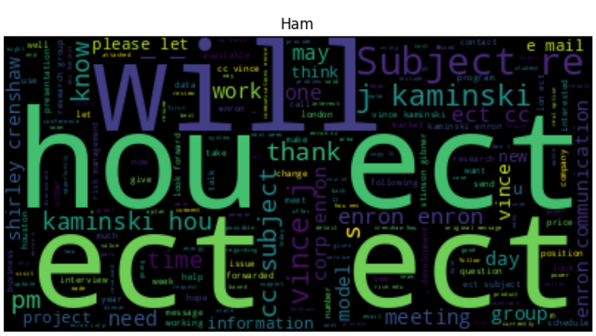

Fig. 5. Word cloud representation of ham (non-spam) emails.

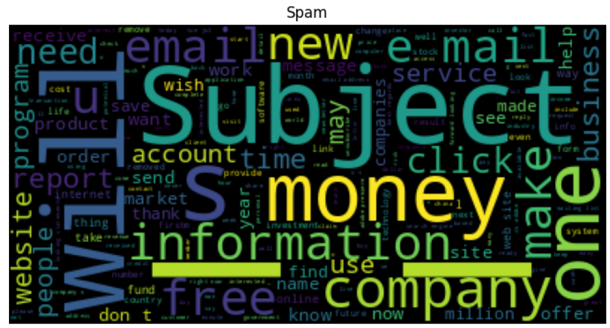

Fig. 6. Word cloud representation of spam emails.

As shown using Matplotlib and WordCloud in Fig. 5, ham emails are dominated by conversational and contextual words such as *hou*, *ect*, *subject*, and names, indicating regular communication patterns.

In contrast, Fig. 6 highlights promotional and action-driven terms such as *free*, *money*, *click*, *offer*, and *company*. These words reflect typical spam strategies involving urgency, rewards, and calls to action.

Overall, the clear distinction in vocabulary between ham and spam emails reinforces the effectiveness of textual features in classification tasks.

G. Prediction Confidence

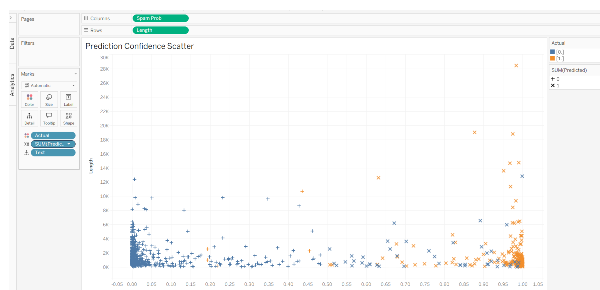

Fig. 7. Prediction confidence scatter plot.

Most predictions cluster near the extreme probability values, further confirming that the model rarely produces uncertain outputs.

H. Feature Analysis

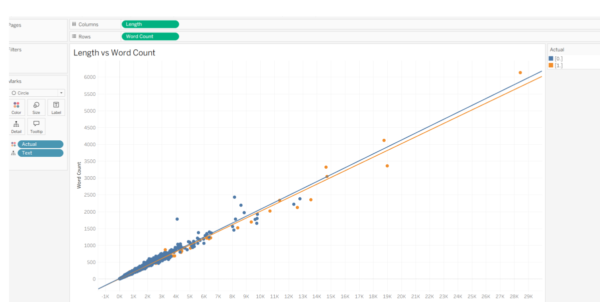

Fig. 8. Message length vs. word count (Tableau visualization).

A strong linear relationship between message length and word count (Fig. 8) validates the internal consistency of the textual features extracted from the dataset.

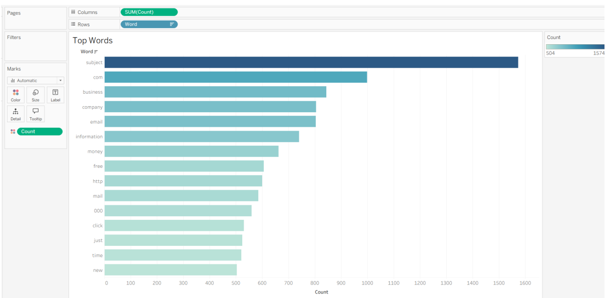

Fig. 9. Top frequent words in spam emails (Tableau visualization).

Words such as *free*, *money*, *click*, and *offer* emerge as strong spam indicators (Fig. 9), consistent with known phishing and marketing spam lexicons.

I. Overall Insights

- Spam emails exhibit distinct linguistic and structural patterns identifiable through EDA.
- Class imbalance significantly impacts model performance; weighted loss functions are essential.
- Feature engineering (BERT + numerical) contributes more to performance than hyperparameter tuning.
- Visualization — both code-based and Tableau — improves interpretability of model behaviour and dataset characteristics.

IX. Conclusion

This study demonstrates the effectiveness of combining data analysis, visualization, and machine learning for spam email detection. Through comprehensive EDA, key patterns were identified including class imbalance, distinctive keyword usage, and variations in message length between spam and ham emails. Visualization techniques — using Matplotlib, Scikit-learn plotting utilities, and Tableau — played a central role in uncovering these patterns.

A hybrid feature representation combining BERT-based text embeddings (frozen, inference-only) and structured numerical features was used with an MLP model trained using a weighted binary cross-entropy loss. The model achieved 96.9% accuracy and an F1-score of 0.951, confirming that the extracted features effectively capture meaningful discriminative patterns.

The study also highlights the importance of addressing class imbalance: an initial unweighted training run produced noticeably lower F1 and ROC-AUC, while the weighted version improved minority-class sensitivity substantially.

Overall, the results indicate that data understanding and feature engineering are more critical than complex model design for this task. The integration of visualization and machine learning provides a robust and interpretable framework for real-world spam detection.

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